Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Automated Brain Tumor Detection and Classification using Deep Learning

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1. Problem Statement

Brain tumors are a serious and life-threatening condition. If doctors can detect them early, patients have a better chance of getting the correct treatment and improving survival. At present, diagnosis mainly depends on radiologists who manually examine Magnetic Resonance Imaging (MRI) scans. This manual process is time-consuming and requires a lot of effort, especially when there are many patients and a large number of scans.

Manual checking can also lead to human error. Different radiologists may sometimes give different opinions, especially when tumor types look similar, such as *Glioma*, *Meningioma*, and *Pituitary* tumors. With the rapid growth of medical imaging data, the workload on radiologists increases, and this can delay diagnosis.

Therefore, there is a clear need for an automated **Computer-Aided Diagnosis (CAD)** system that can support radiologists by providing a fast, consistent, and accurate "second opinion." This project aims to develop a deep learning-based system using **Convolutional Neural Networks (CNNs)** to automatically detect and classify brain tumors from MRI images with high precision.

How to use references using BibTeX files [1–3].

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2. Objectives

- To design and implement an automated system to detect brain tumors from MRI images using image processing techniques.
- To classify detected tumors into types such as *Glioma*, *Meningioma*, and *Pituitary* using deep learning models like **CNNs**.
- To preprocess MRI images using noise removal, skull stripping, and normalization to improve model performance.
- To compare different CNN architectures (e.g., **VGG16**, **ResNet50**, **EfficientNet**) using accuracy, precision, and recall.
- To develop a user-friendly tool or interface that can assist medical professionals during diagnosis.

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3. Literature Review

Brain tumor classification is an important task in medical image analysis because correct diagnosis helps doctors choose the best treatment. **MRI** is widely used because it shows soft tissues clearly. However, manual interpretation of MRI scans is slow, and results can vary between observers. Because of this, many researchers have developed automated systems using **Computer Vision (CV)**, **Machine Learning (ML)**, and **Deep Learning (DL)**.

This section summarizes important work in the area, starting from traditional machine learning methods and moving to modern deep learning approaches.

3.1. Traditional Machine Learning Approaches

Early research mainly followed a classical pipeline: preprocessing, manual feature extraction, and then classification. Many studies used feature extraction methods such as **Discrete Wavelet Transform (DWT)** and texture analysis using **Gray Level Co-occurrence Matrix (GLCM)**. After extracting features, classifiers such as **Support Vector Machines (SVM)** were commonly used. The reference [1] provides strong background knowledge for many of these image processing methods. However, seminal work by Pereira et al. [4] demonstrated that CNNs could learn these features automatically with higher accuracy.

Although these traditional methods worked reasonably well on small datasets, they had limitations. Their performance depended heavily on hand-crafted features and the chosen feature set. As a result, they often struggled to generalize to new scans and could not easily capture complex tumor patterns. However, recent hybrid approaches have successfully combined these traditional features with deep learning, such as the Deep Wavelet Auto-Encoder model proposed by [5], which utilizes wavelet functions within a neural network architecture.

3.2. The Rise of Deep Learning and Convolutional Neural Networks (CNNs)

Deep learning, especially **CNNs**, improved brain tumor classification by learning features directly from images. This reduces the need for manual feature engineering and supports end-to-end learning.

Study 1: Simple CNN Architectures

[6] proposed a simple CNN model to classify tumors into *Glioma*, *Meningioma*, and *Pituitary*. They used around 3064 MRI slices.

- **Methodology:** A custom CNN was trained from scratch using convolutional layers, max pooling, and dense layers.
- **Results:** High training accuracy was reported, but validation accuracy was lower, showing overfitting.

- **Strengths:** Demonstrated that CNNs can learn features automatically and can outperform traditional ML methods.
- **Limitations:** Small datasets can lead to overfitting without enough augmentation and regularization.

Study 2: Transfer Learning and Fine-tuning
To reduce the small dataset problem, [2] used **Transfer Learning** with pre-trained models (e.g., VGG19, AlexNet, ResNet).

- **Methodology:** They used pre-trained CNNs and applied fine-tuning strategies to adapt the model to MRI images.
- **Results:** Their VGG19-based approach achieved strong accuracy (about 94.82% in cross-validation).
- **Strengths:** Transfer learning improves generalization and training stability for small medical datasets.
- **Limitations:** VGG models are heavy and computationally expensive.

Study 3: Deep Transfer Learning with GoogLeNet
[7] explored **GoogLeNet** using Inception modules.

- **Methodology:** Features were extracted using GoogLeNet and classified using Softmax and other classifiers.
- **Results:** High classification performance was reported on common datasets.
- **Strengths:** Inception modules capture features at multiple scales, useful because tumors vary in size.
- **Limitations:** Some hybrid approaches add complexity compared to pure end-to-end CNNs.

3.3. Advanced Architectures and Data Augmentation

As the field developed, researchers tested advanced models and methods to improve performance and reduce common issues such as overfitting and loss of spatial information.
Study 4: Deep Wavelet Auto-Encoders
[5] introduced a **Deep Wavelet Auto-Encoder (DWAE)** model.

- **Methodology:** Combined an auto-encoder with wavelet activation functions and utilized a seed-growing segmentation approach.
- **Results:** Achieved a high accuracy of 99.3% on BRATS datasets.
- **Strengths:** Effectively handles noise and heterogeneity in MRI images through its specialized filtering and encoding process.

- **Limitations:** Increased model complexity and specialized architecture may require more computational resources.

Study 5: Capsule Networks (CapsNets)
[3] proposed **Capsule Networks** to preserve spatial information that can be lost due to pooling in CNNs.

- **Methodology:** They used capsule layers and included tumor boundaries to guide learning.
- **Results:** The CapsNet achieved decent accuracy and improved robustness in some cases.
- **Strengths:** Better at capturing spatial relationships and pose information.
- **Limitations:** Training can be unstable and computationally expensive.

Study 6: Data Augmentation Importance
[8] showed that **Data Augmentation** (rotation, flipping, zooming) helps reduce overfitting and improves performance.

- **Methodology:** They used strong augmentation and fine-tuned a deep CNN.
- **Results:** Accuracy improved significantly with augmentation compared to no augmentation.
- **Significance:** Augmentation is very important for small and imbalanced medical datasets.

3.4. Summary of Recent Reviews

A review paper by [9] discussed major challenges in brain tumor classification using deep learning:

1. **Data Imbalance:** Some tumor classes have more images than others.
2. **Explainability:** Deep learning models can behave like black boxes, reducing trust.
3. **Generalizability:** Models trained on one hospital or scanner may not work well on another.

3.5. Research Gap and Proposed Contribution

Even though many studies report high accuracy, there are still gaps that our project aims to address:

- **Gap 1: Computational Efficiency.** High-accuracy models like VGG19 [2] are heavy. There is a need for efficient architectures like **ResNet50** and **EfficientNet**.

- **Gap 2: End-to-End Simplicity.** Complex hybrid pipelines increase implementation effort. We aim for a clean and simple end-to-end workflow.
- **Gap 3: Reproducibility on Public Data.** We will focus on public datasets such as Kaggle and BraTS to support reproducibility.

In this project, we plan to implement a streamlined **Transfer Learning** pipeline using a modern backbone (such as ResNet50 or EfficientNet). We expect that strong preprocessing (including skull stripping) and strong data augmentation can help achieve high accuracy while keeping the model more efficient than heavier networks.

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4. Methodology

The proposed methodology follows a standard deep learning workflow for medical image analysis. It is divided into four phases: **Data Acquisition & Preprocessing**, **Data Augmentation**, **Model Architecture & Transfer Learning**, and **Training & Evaluation**.

4.1. Phase 1: Data Acquisition and Preprocessing

MRI images can contain noise, artifacts, and non-brain tissues that can reduce model performance. Therefore, preprocessing is important.

1. **Dataset Selection:** We will use the **Kaggle Brain Tumor MRI Dataset** and possibly the **BraTS** dataset. The dataset includes four classes: *Glioma*, *Meningioma*, *Pituitary*, and *No Tumor*.

2. **Grayscale Conversion & Resizing:** MRI images are grayscale, but many pre-trained models require a 3-channel input. We will replicate the grayscale channel three times. Images will be resized to a fixed size (e.g., 224×224) to match model input requirements.

3. **Noise Removal (Denoising):** We will reduce noise using filters:

   - *Gaussian Filter:* Helps smooth high-frequency noise.
   - *Median Filter:* Helps remove noise while preserving edges, which is important for tumor boundaries. This technique has been shown to improve segmentation results in deep learning models, such as the DWAE model implemented by [5].

4. **Skull Stripping (Brain Extraction):** Skull and surrounding tissues do not help classification. We will use a contour-based method using **OpenCV**:

   - Apply thresholding to create a binary mask.
   - Find the largest contour corresponding to the brain.
   - Crop using bounding box or extreme points to remove unwanted regions.

5. **Normalization:** We will apply **Min-Max Normalization** to scale pixel values to $[0, 1]$ for stable training and faster convergence.

4.2. Phase 2: Data Augmentation

Deep learning models can overfit when data is limited. To reduce overfitting, we will use data augmentation during training.

- **Random Rotation:** ± 10 to 30 degrees.
- **Horizontal Flip:** Mirror images to increase variation.

- **Brightness/Contrast adjustment:** Simulate different scanner settings.
- **Zoom/Shift:** Small zoom and translations to improve robustness.

4.3. Phase 3: Deep Learning Model Architecture

We will compare two approaches: a Custom CNN and a Transfer Learning model.

4.3.1 Approach A: Custom CNN

We will build a lightweight CNN model inspired by architectures such as the one proposed by Badža et al. [10], which showed that simpler networks can be effective for tumor classification:

- **Convolutional Layers** to learn features such as edges, textures, and tumor shapes.
- **ReLU Activation** to introduce non-linearity.
- **Max-Pooling Layers** to reduce size and computation.
- **Fully Connected Layers** for final classification.
- **Dropout** (e.g., 50%) to reduce overfitting.

4.3.2 Approach B: Transfer Learning (e.g., ResNet50 / VGG16 / EfficientNet)

This will be the main approach because it performs well with small datasets. We will utilize powerful backbones such as ResNet50, originally introduced by He et al. [11] and EfficientNet [12], which optimizes accuracy and computational efficiency.

- **Backbone Feature Extractor:** Use a pre-trained model trained on ImageNet.
- **Freezing Layers:** Freeze early layers at first to keep general feature detectors.
- **Custom Head:** Add a new head (Global Average Pooling + Dense + Softmax) for 4 classes.
- **Fine-Tuning:** Unfreeze the last blocks and fine-tune with a low learning rate.

4.4. Phase 4: Training and Evaluation Strategy

We will implement the system in **Python** using **TensorFlow/Keras**.

1. **Dataset Split:**

- **Training Set (70%):** Used to train model weights.
- **Validation Set (15%):** Used to tune hyperparameters and monitor overfitting.
- **Test Set (15%):** Used for final performance evaluation.

2. Hyperparameters:

- **Optimizer:** Adam
- **Loss Function:** Categorical Cross-Entropy
- **Batch Size:** 32 or 64 (depending on GPU memory)
- **Epochs:** 20–50 with **Early Stopping** (patience of 5 epochs)

3. Performance Metrics:

- **Accuracy**
- **Precision**
- **Recall (Sensitivity)**
- **F1-Score**
- **Confusion Matrix**

Tools and Libraries:

- **OpenCV (cv2)**
- **TensorFlow / Keras**
- **Scikit-Learn**
- **Matplotlib/Seaborn**

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5. Dataset / Data Collection

We will use the publicly available **Kaggle Brain MRI Dataset** and possibly the **BraTS (Multimodal Brain Tumor Segmentation Challenge)** dataset. The Kaggle dataset usually contains over 3,000 MRI images categorized into four classes: *Glioma*, *Meningioma*, *Pituitary tumor*, and *No Tumor*. The dataset may include different MRI modalities such as **T1**, **T2**, and **FLAIR**.

To ensure good quality, we will preprocess the images to reduce noise, remove unwanted regions, and standardize image size. Ethical use is ensured because these are anonymized and open-access datasets intended for research purposes.

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6. Expected Outcomes

- A working deep learning model that can classify brain MRI images with an accuracy above 90%.
- A performance report including Confusion Matrices, Precision, Recall, and F1-scores for each tumor class.
- A comparison report showing strengths and weaknesses of different CNN models (e.g., ResNet vs VGG vs EfficientNet).
- A demonstration of correct predictions on unseen test images, showing the system's practical usefulness.

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7. Percentage of Image Processing & Computer Vision Involvement

This project is almost 100% based on **Image Processing** and **Computer Vision**. The main workflow includes:

1. **Image Preprocessing:** Noise removal using Gaussian/Median filters, skull stripping, normalization, and possible contrast enhancement.
2. **Feature Extraction & Classification:** Using **CNNs** and **Transfer Learning** models to learn visual features directly from MRI images.

There are no external hardware or non-CV components. The entire solution depends on analyzing MRI image data.

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